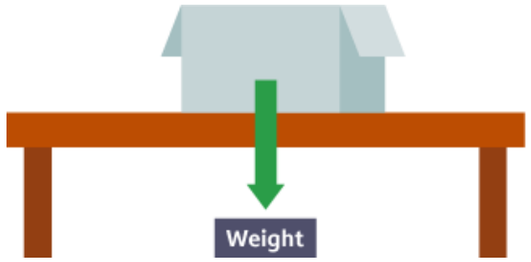
Knowledge Organiser
Year 7: Forces – Mass and Weight

What is mass?
- Mass is a physical quantity
- It is a measure of the **amount of matter** in an object
- If an object has more particles it will have a higher mass than it did before
- The standard units of mass are **kilograms, kg**

How is weight different?
- Weight **is a force** caused by the pull on an object due to gravity
- Like other forces the standard units of weight are **Newtons, N**
- If the acceleration on an object due to gravity is increased, then its weight will be higher



A force diagram showing the weight acting on a box on a table

Weight and other planets
- Weight is a result of gravity interacting with mass
- An object's weight will change depending on the strength of the gravitational field it is in
- If an object has a weight of 10N on Earth, it will weight around 4N on Mars

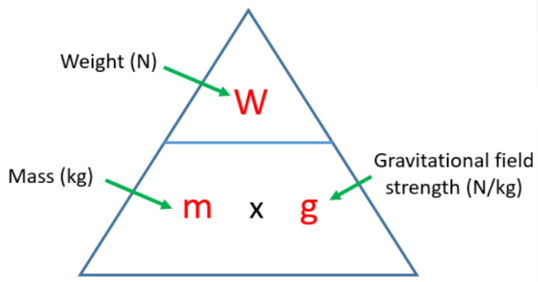
Planet	Gravitational field strength (N/kg)
Mercury	4
Venus	9
Earth	10
Mars	4
Jupiter	23
Saturn	9
Uranus	9
Neptune	11



A labelled diagram showing how a person's mass and weight would be different on the moon than Earth

Doesn't mass change as well?
- Mass is determined by the number of particles in an object
- Unless the object loses or gains matter, its mass will stay the same
- So you would have the same mass regardless of which planet you are on but your weight would change

A formula triangle for the equation used to calculate weight



How is weight calculated?
- Weight is equal to the mass of an object multiplied by the gravitational field strength acting on it
- The gravitational field strength on Earth is usually rounded to 10 N/kg (more accurarely 9.81 N/kg)
- This means that an object with a mass of 5kg would have a weight of 5 kg x 10 N/kg = 50N

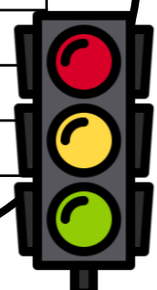


Standard units?
- A physical quantity is something which can be measured about an object
- Each physical quantity, has its own **standard unit** e.g. kilograms for mass
- These units are used for that object unless otherwise stated

Quantity	Symbol	Standard Units	Symbol for unit
Time	t	Seconds	s
Mass	M	Kilograms	kg
Volume	V	Centimetres cubed	cm^3
Force	F	Newtons	N
Temperature	T	Kelvin	K
Distance	d	Metres	m

RAG Yourself!

Learning Outcomes;	RAG
State what is meant by a resultant forces	
Calculate resultant forces on an object	
Describe what is meant by balanced forces	
Draw force diagrams using arrows	
State the units of mass and weight	
Describe the factors which affect mass and weight	
Calculate mass and weight using W = m x g	



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